

THE ZERO PARADOX

ZP-J: Executability of Self-Reference

Version 1.2 | April 2026

v1.2: Minor wording fix. | v1.1: Remark R-J.0 added — Lean encoding scope. | v1.0: Initial release — Theorem T-EXEC; all ZPJ.lean theorems axiom-free.

This document establishes Theorem T-EXEC (Executability of Self-Reference): in any ZP-A join-semilattice with AFA (Anti-Foundation Axiom) grounding, the unique Quine atom Q — the element satisfying $Q = \{Q\}$ — is provably the bottom element $\perp$. This closes the bridge between the set-theoretic and order-theoretic layers of the framework. CC-1 from ZP-A, which stated " $S_0 = \perp$ " as a modelling commitment, becomes a derived theorem.

The key is a single structural field in the AFAStructure typeclass: `bot_self_mem`, which encodes that the bottom element is self-containing. With this, the proof of T-EXEC is three lines. No bridge axiom. No freestanding commitment. The identification $\perp = \{\perp\}$ — implicit in the framework since ZP-E's DA-1 Path 1 — is now a verified structural prerequisite. (See Remark R-J.0 for the precise scope of this verification: the Lean proof encodes AFA as a typeclass field, not as a ZF+AFA set-theoretic derivation.)

Section I: The Open Question — CC-1 as a Modelling Commitment

I. CC-1 in ZP-A

ZP-A Conditional Claim CC-1 states: if the state sequence is initialised at $\perp$, then $\perp \leq S(n)$ for all n . This follows trivially from T2 ($\perp$ is the global minimum), so its formal content is essentially: we are choosing $\perp$ as the initial state S_0 .

The label "Conditional Claim" (CC) marks this as a modelling commitment — an explicit choice not derivable from the axioms A1–A4 alone. ZP-A's axioms give us a semilattice with a bottom element. They do not say which instantiation of the semilattice should start at that bottom. CC-1 asserts: ours does. This is well-motivated, but it is a choice.

The question ZP-J investigates: is this choice forced? Is there a structural reason — derivable from the framework's foundational commitments — that any well-grounded instantiation of ZP-A must begin at $\perp$? The answer is yes, provided the lattice has AFA grounding.

II. The Implicit Identification

ZP-E's DA-1 Path 1 already states: $\perp = \{\perp\}$ (Quine atom, ZF+AFA). This identification — the bottom element is the unique self-containing set under AFA — was present informally but never formally bridged to CC-1. It appeared as motivation for why $\perp$ is the right starting point, not as a derivation of it.

The gap: ZP-A's lattice order is abstract (defined by axioms A1–A4). AFA is a set-theoretic axiom. There is no automatic connection between "x is self-containing" and "x is the lattice bottom." Connecting them requires a bridge — and that bridge was the missing piece. ZP-J provides it.

Open question entering ZP-J: Is CC-1 ($S_0 = \perp$) forced by the framework's foundational structure, or is it an independent modelling choice? If forced, what is the structural reason? Answer (ZP-J T-EXEC): it is forced — by the AFA identification $\perp = \{\perp\}$, encoded structurally in the `AFAStructure` typeclass.

Section II: AFA Machinery — Self-Membership and the Quine Atom

I. The Anti-Foundation Axiom

Standard set theory (ZF) includes the Foundation Axiom: every non-empty set S contains an element disjoint from S . A consequence is that no set can contain itself: $x \in x$ is impossible in ZF. This rules out self-referential sets by fiat.

The Anti-Foundation Axiom (AFA, Aczel 1988) replaces Foundation with a universal existence and uniqueness result: every accessible pointed graph (APG) has a unique set-theoretic decoration. Under AFA, self-containing sets are not only possible but uniquely characterised. The resulting theory ZF+AFA is consistent and expressive — it is the natural set-theoretic home for fixed-point and self-referential structures.

II. The Quine Atom

Under AFA, the equation $x = \{x\}$ has a unique solution. This solution is called the Quine atom, denoted Q . Q is the unique set that contains itself as its sole member: $Q = \{Q\}$. It is not the empty set ($\emptyset = \{\}$ contains nothing); it is not any well-founded set (those cannot contain themselves by Foundation's analogue in the well-founded universe). Q is the minimal non-trivial AFA set — it contains exactly one thing, and that thing is itself.

The AFA uniqueness guarantee is the key property: there is at most one Quine atom. Any two self-containing elements are equal. This means the self-membership property uniquely identifies an element — a "fingerprint" that belongs to exactly one object in the universe.

AFA Uniqueness (`AFAStructure.quine_unique`)

For any type L with AFA structure: if $x, y \in L$ both satisfy `selfMem(x)` and `selfMem(y)`, then $x = y$.

Informally: the Quine atom is unique. Self-containment is a property held by at most one element of any AFA-structured type.

Lean: `AFAStructure.quine_unique` — encoded as a typeclass field, not a theorem, because AFA is a foundational axiom and cannot be derived from type-theoretic principles alone. It is a structural prerequisite for any AFA-grounded lattice.

III. Self-Membership as a Lattice Predicate

In ZP-J, the AFA machinery is abstracted minimally. We do not need the full apparatus of accessible pointed graphs, bisimulation, or set decoration. We need only two things: a predicate `selfMem` on the lattice elements, and the guarantee that it is held by at most one element (`quine_unique`). The third structural field — `bot_self_mem` — is where the bridge is built.

The `AFAStructure` typeclass in `ZPJ.lean` captures exactly this. Its three fields are sufficient to derive T-EXEC with no additional axioms. The full AFA machinery (APGs, bisimulation, decoration) provides the informal justification for why any concrete lattice grounded in ZF+AFA satisfies these three fields — but the formal derivation requires only the fields themselves.

Section III: AFAStructure — The Structural Bridge

I. The Three Fields

The `AFAStructure` typeclass for a ZP-A semilattice L has three fields. The first two encode standard AFA properties. The third is the bridge.

AFAStructure Typeclass (ZPJ.lean § I)

`class AFAStructure (L : Type*) [ZPSemilattice L] with:`

(1) `selfMem : L → Prop` — the self-membership predicate. `selfMem(x)` means x contains itself as a member in the AFA sense.

(2) `quine_unique : ∀ x y : L, selfMem(x) → selfMem(y) → x = y` — AFA uniqueness. At most one element of L is self-containing.

(3) `bot_self_mem : selfMem(⊥)` — the bridge field. The bottom element of the lattice is self-containing. This is the formal encoding of $\perp = \{\perp\}$.

II. What `bot_self_mem` Says

`bot_self_mem` is the single structural claim that connects the order-theoretic world (ZP-A's lattice) to the set-theoretic world (AFA). It says: the bottom element $\perp$ of the lattice is self-containing — it satisfies `selfMem(⊥)`.

In set-theoretic terms: $\perp \in \perp$, i.e. $\perp = \{\perp\}$. In the framework: the null state contains itself as its sole content. This is not a new claim — it is the identification that ZP-E's DA-1 Path 1 already invokes informally. ZP-J encodes it as a typeclass field, making it a verifiable structural prerequisite rather than a narrative motivation.

Any concrete lattice L that claims to be AFA-grounded must prove `bot_self_mem` as part of its `AFAStructure` instance. If it cannot, it is not genuinely AFA-grounded — the identification $\perp = \{\perp\}$ is part of what "AFA-grounded" means.

Remark R-J.0 — CIC Encoding and the AFA Distinction

In the `MachinePhase` concrete instance (ZP-K), `selfMem` is defined as `selfMem x := x = ⊥`. With this definition, `bot_self_mem` is proved by `rfl`: $\perp = \perp$ holds by reflexivity.

Remark R-J.0 — CIC Encoding and the AFA Distinction

This is the CIC-compatible encoding of AFA self-containment. In Lean 4 (based on the Calculus of Inductive Constructions), the set-theoretic statement $\perp \in \perp$ — meaning $\perp$ literally contains itself as a member under ZF+AFA — is not directly formalizable. CIC lacks the membership relation $\in$ of set theory. The encoding `selfMem x := x =  $\perp$` captures the structural role: "self-containing" means "equals the bottom element." The Lean proof compiles by `rfl` because self-containing is defined as equality with $\perp$.

This is a structural analogy, not a set-theoretic derivation from ZF+AFA. The full set-theoretic content of AFA — that $\perp$ literally contains itself as a member of the AFA universe — is not present in the Lean proof. What the typeclass encodes is the structural consequence: there is a unique element playing the Quine role, and $\perp$ is that element. The AFA grounding provides the informal justification for why this structural role belongs to $\perp$; the Lean proof requires only the typeclass fields, not the full set-theoretic apparatus of ZF+AFA.

III. Why a Typeclass Field Rather than a Freestanding Axiom

In the stub version of `ZPJ.lean` (commit 629a534), the bridge was a freestanding axiom: `ax_j1_quine_join_identity`. It stated directly that the Quine atom satisfies the join-identity. This compiled, but the purity check showed T-EXEC depending on `ax_j1_quine_join_identity` — a named axiom floating outside any typeclass.

Encoding the commitment as a typeclass field is philosophically and formally cleaner. A freestanding axiom is a global assertion that the proof checker accepts without verification. A typeclass field is a requirement: any instance of `AFAStructure` must supply a proof of `bot_self_mem`. The commitment is not asserted once and forgotten — it is checked at every instantiation. Concrete lattices must earn their AFA status.

The distinction: a freestanding axiom says "trust me, this is true." A typeclass field says "prove it for your specific lattice, or it does not compile." ZP-J uses the second. The philosophical commitment ($\perp = \{\perp\}$) becomes a proof obligation at every concrete instantiation.

Section IV: Theorem T-EXEC — The Quine Atom is Bottom

Theorem T-EXEC — Executability of Self-Reference

Statement: Let L be a ZP-A semilattice with `AFAStructure`. If $q : L$ is a Quine atom (`selfMem(q)` and q is unique among self-containing elements), then $q = \perp$.

Lean: `ZeroParadox.ZPJ.t_exec` — proved in `ZPJ.lean`. Purity: does not depend on any axioms. ✓

I. The Proof

The proof of T-EXEC is immediate from the typeclass fields. It has three steps:

- `hq.2` states: every self-containing element of L equals q . (This is the uniqueness half of `IsQuineAtom q`.)

- `AFAStructure.bot_self_mem` states: $\perp$ is self-containing. (This is the bridge field — $\perp = \{\perp\}$ encoded structurally.)
- Applying `hq.2` to $\perp$ using `bot_self_mem` gives: $\perp = q$. By symmetry: $q = \perp$. QED.

The entire proof is one line in Lean 4: `(hq.2 bot AFAStructure.bot_self_mem).symm`. No appeal to the join operation. No bridge axiom. No DA-2. Just AFA uniqueness applied at $\perp$.

Remark R-J.1

The proof does not use `da2_bottom_characterization` (from ZP-E). That result — “ $(\forall x, \text{join } S \ x = x) \leftrightarrow S = \perp$ ” — is used in the derived theorem J1 (Section V), but T-EXEC itself is purely order-theoretic: it uses only `quine_unique` and `bot_self_mem`.

II. IsQuineAtom bot

A corollary of T-EXEC is that $\perp$ itself is a Quine atom. This is the converse direction: not only does the Quine atom equal $\perp$ (T-EXEC), but $\perp$ equals the Quine atom (`bot_is_quine_atom`).

Proposition — bot_is_quine_atom

In any `AFAStructure` lattice `L`: `IsQuineAtom( $\perp$ )`.

Proof: $\perp$ is self-containing by `bot_self_mem`. Any other self-containing `x` satisfies $x = \perp$ by `quine_unique(x,  $\perp$ , selfMem(x), bot_self_mem)`.

Lean: `ZeroParadox.ZPJ.bot_is_quine_atom` — does not depend on any axioms. ✓

III. The Full Biconditional

Combining T-EXEC and `bot_is_quine_atom` yields the full biconditional: `IsQuineAtom(q)  $\leftrightarrow$  q =  $\perp$` . Being the Quine atom and being the bottom element are the same property, stated in set-theoretic and order-theoretic language respectively.

Theorem t_exec_iff — Full Equivalence

For any `q : L` in an `AFAStructure` lattice:

`IsQuineAtom(q)  $\leftrightarrow$  q =  $\perp \leftrightarrow \forall x : L, \text{join } q \ x = x$` .

The three conditions are mutually equivalent: Quine atom (set-theoretic), bottom element (order-theoretic), and join-identity element (algebraic). They are three formulations of one structural role.

Lean: `ZeroParadox.ZPJ.t_exec_iff` — does not depend on any axioms. ✓

Section V: Derived Results — J1 and CC-1 as Theorems

I. J1 — QuineJoinIdentity (Derived)

In the initial stub of ZPJ.lean, the claim "the Quine atom satisfies the join-identity" was stated as a freestanding axiom (ax_j1_quine_join_identity). ZP-J replaces it with a theorem.

Theorem J1 — QuineJoinIdentity (formerly Axiom AX-J1)

In any AFAStructure lattice L , if q is the Quine atom, then $\forall x : L, \text{join } q \ x = x$.

Proof: $q = \perp$ by T-EXEC. Then $\text{join } q \ x = \text{join } \perp \ x = x$ by A4 (bot_join, ZP-A). ✓

Status: DERIVED THEOREM. Was axiom ax_j1_quine_join_identity in the stub — now proved from T-EXEC + ZP-A A4. No freestanding axiom remains.

Lean: ZeroParadox.ZPJ.j1_quine_join_identity — does not depend on any axioms. ✓

The elimination of AX-J1 is the payoff of bot_self_mem. In the stub, we could prove T-EXEC from AX-J1 (via da2_bottom_characterization), but AX-J1 was itself an axiom — the buck stopped there. With bot_self_mem, the buck stops at the typeclass definition: a structural requirement, not a global assertion.

II. CC-1 Derived

ZP-A CC-1 is now a theorem. If the initial state S_0 of a state sequence is the Quine atom Q , then $S_0 = \perp$ — a consequence of T-EXEC, not a modelling choice.

Theorem CC-1 (Derived) — cc1_derived

Let L be an AFAStructure lattice. Let $S : \mathbb{N} \rightarrow L$ be a state sequence (ZP-A D3) and $Q : L$ a Quine atom. If $S(0) = Q$, then $S(0) = \perp$.

Proof: $S(0) = Q$ (hypothesis). $Q = \perp$ by T-EXEC. Therefore $S(0) = \perp$.

The modelling commitment "we choose $\perp$ as the initial state" is replaced by "the Quine atom IS $\perp$, structurally." Starting at Q and starting at $\perp$ are not two choices — they are the same state, identified by structure.

Lean: ZeroParadox.ZPJ.cc1_derived — does not depend on any axioms. ✓

III. Uniqueness of the Bottom Role

A further consequence is algebraic uniqueness: in any ZP-A semilattice (without even requiring AFA structure), at most one element can satisfy the join-identity. This is a clean corollary of da2_bottom_characterization (ZP-E).

Theorem — bot_unique

For any ZP-A semilattice L (no AFA required): if $x, y : L$ both satisfy $\forall z, \text{join } x \ z = z$ and $\forall z, \text{join } y \ z = z$, then $x = y$.

Proof: da2_bottom_characterization gives $x = \perp$ and $y = \perp$. Therefore $x = y$.

Lean: ZeroParadox.ZPJ.bot_unique — does not depend on any axioms. ✓

Section VI: Implications — What Was the Commitment?

I. The Remaining Foundation

T-EXEC and its corollaries carry zero freestanding axioms in the Lean 4 purity check. The entire derivation traces to the fields of two typeclasses: ZPSemilattice (from ZP-A) and AFAStructure (new in ZP-J). No standalone axiom statement appears anywhere in the ZPJ.lean proof obligations.

The foundational commitments are the typeclass fields themselves. In ZPSemilattice: A1–A4 (the semilattice axioms). In AFAStructure: selfMem (a predicate), quine_unique (AFA uniqueness), and bot_self_mem (the bridge). These are the structural prerequisites for any lattice that claims ZP-A + AFA grounding. They are not axioms floating in the ambient theory — they are proof obligations that any concrete instance must discharge.

II. Where the Philosophy Lives Now

The philosophical content of $\perp = \{\perp\}$ has not disappeared. It has moved. Previously it lived in a narrative comment in ZP-A (CC-1's informal motivation) and in ZP-E's DA-1 Path 1 (one of three informal arguments for instantiation as execution). Now it lives in the definition of AFAStructure itself — specifically in bot_self_mem.

This is the correct location for a foundational claim. It is not hidden in a proof; it is stated in the typeclass definition where any reader can see it. Any lattice that wants to use ZP-J's results must provide a proof that its bottom element is self-containing. If it cannot, it is simply not an AFA-grounded lattice in the sense required by this framework.

III. The Formal Chain is Now Closed

The derivation chain entering ZP-J had one remaining gap: CC-1 was a committed starting point, not a derived one. The chain from AFA structure to T-SNAP now has no such gaps. Every node is either a proved theorem or a typeclass field:

- ZP-A axioms A1–A4: semilattice structure (ZPSemilattice fields).
- ZP-B: 2-adic topology and irreversibility (proved from Mathlib).
- ZP-C: information theory, L-INF, L-RUN (proved axiom-free or from Mathlib).
- ZP-D: state layer, orthogonality (proved from ZP-A and ZP-B).
- ZP-E: T-SNAP, DA-1, DA-2 (T-SNAP and DA-2 proved axiom-free; DA-1 Path 3 outside Lean scope).
- ZP-J: AFAStructure fields (selfMem, quine_unique, bot_self_mem). T-EXEC, J1, CC-1: proved axiom-free. ✓

DA-1 Path 3 (the AIT/Kolmogorov complexity argument in ZP-E) remains outside Lean scope for the same reason it always has: Kolmogorov complexity is uncomputable and absent from Mathlib. This is a Lean tooling limitation, not a gap in the mathematics. ZP-J does not affect DA-1's status.

Key Results — ZP-J v1.1

T-EXEC: In any AFAStructure lattice, $\text{IsQuineAtom}(q) \leftrightarrow q = \perp$. Self-reference and bottom-hood are the same structural role.

J1 (Derived): The Quine atom satisfies the join-identity $\forall x, \text{join } q \ x = x$. Was axiom ax_j1 in the stub — now a theorem from T-EXEC + A4.

Key Results — ZP-J v1.1

CC-1 (Derived): $S_0 = Q \Rightarrow S_0 = \perp$. The modelling commitment is replaced by a structural consequence.

Lean purity: all ZPJ.lean theorems verify "does not depend on any axioms" ✓ (modulo AFAStructure typeclass fields quine_unique and bot_self_mem, which carry the same logical status as axioms but are not surfaced by #print axioms).

Traceability Register — ZP-J v1.1

Claim	Grounded In	New axiom?	Status
T-EXEC: Quine atom = $\perp$	AFAStructure.quine_unique + AFAStructure.bot_self_mem	None — typeclass fields, not freestanding axioms	Lean: t_exec — does not depend on any axioms ✓
J1: Quine join-identity	T-EXEC + ZP-A A4 (bot_join)	None	Lean: j1_quine_join_identity — axiom-free ✓ (was axiom ax_j1 in stub)
CC-1 (Derived): $S_0 = Q \Rightarrow S_0 = \perp$	T-EXEC	None	Lean: cc1_derived — axiom-free ✓ (was ZP-A conditional claim)
bot_is_quine_atom: IsQuineAtom($\perp$)	AFAStructure.bot_self_mem + quine_unique	None	Lean: bot_is_quine_atom — axiom-free ✓
t_exec_iff: IsQuineAtom(q) $\leftrightarrow$ q = $\perp$	T-EXEC + bot_is_quine_atom	None	Lean: t_exec_iff — axiom-free ✓
bot_unique: join-identity is unique	ZPE.da2_bottom_characterization	None	Lean: bot_unique — axiom-free ✓ (no AFA required)
AFAStructure.bot_self_mem	AFA (ZF+AFA) — $\perp = \{\perp\}$	Typeclass field — proof obligation at instantiation	Not a theorem; a structural prerequisite. Must be discharged by each concrete instance.

Open Items Register — ZP-J v1.1

Item	Status	Description
CC-1 (ZP-A) derivability	CLOSED — T-EXEC (ZP-J)	CC-1 is now a theorem. The Quine atom = $\perp$ is structurally derived. No freestanding axiom. No modelling commitment beyond AFAStructure typeclass fields.

Item	Status	Description
AX-J1 bridge axiom	CLOSED — J1 derived	The stub version of ZPJ.lean had ax_j1 as a freestanding axiom. The final version derives J1 as a theorem from T-EXEC + A4. Axiom eliminated.
AFAStructure concrete instances	CLOSED — ZP-K MachinePhase	ZP-K provides machinePhaseAFA : AFAStructure MachinePhase, discharging bot_self_mem for ZP-E's two-element machine. selfMem x := x = $\perp$; bot_self_mem := rfl. The Q_2 model (ZP-B) is a natural extension but not required for T-SNAP or DA-1.
DA-1 Path 1 formalisation	OPEN — outside Lean scope	DA-1 Path 1 (ZP-E) invokes $\perp = \{\perp\}$ informally. ZP-J now provides the formal grounding for that identification. A future revision of ZP-E could cite ZP-J T-EXEC as the formal basis for Path 1, closing the last informal step in the DA-1 three-path argument.
ZP-A CC-1 label update	OPEN — editorial	ZP-A still labels CC-1 as a Conditional Claim. With T-EXEC established in ZP-J, the label can be updated to Derived Theorem (citing ZP-J). This is an editorial update, not a mathematical change. Other ZP documents are not modified by ZP-J.

End of ZP-J v1.1 | Theorem T-EXEC: Executability of Self-Reference | CC-1 derived — no freestanding axioms | All ZPJ.lean theorems: does not depend on any axioms (modulo AFAStructure fields quine_unique, bot_self_mem — same logical status as axioms; not surfaced by #print axioms) | Remaining foundation: ZPSemilattice (A1–A4) and AFAStructure (selfMem, quine_unique, bot_self_mem)